



**COMPETENCY STANDARD
FOR
MACHINE SHOP PRACTICE
(*LIGHT ENGINEERING SECTOR*)**

**Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance**

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The Competency Standards for **Machine Shop Practice** is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment and subsequently, they would be qualified and settled for a relevant job.

This document was initially developed as part of the **Skills for Employment Investment Program (SEIP)** and has been reviewed and updated for use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)**. Its purpose is to address the skill requirements of the Light Engineering sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical

competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at Glance:

Generic (Basic) Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-MSP-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Practice personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-LE-MSP-02-G	Communicate in English in the Workplace	1. Read, understand and prepare documents in English 2. Prepare workplace documents in English 3. Perform conversations in English language	10
SICIP-LE-MSP-03-G	Operate in a Self-Directed Team	1. Identify team goals and work processes 2. Communicate and cooperate with team members. 3. Work as a team member. 4. Solve problems as a team member	10
Total Hour			30

Sector Specific (Common) Competencies (40 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-MSP-01-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	04
SICIP-LE-MSP-02-S	Translate Manuals, Drawings, Specifications and Estimate	1. Understand symbols, mechanical drawing, specifications and way of estimation 2. Interpret manuals, schematic drawings specific to ship engineering works 3. Store manuals, drawings and plans	12
SICIP-LE-MSP-03-S	Work with Hand Tools, Power Tools	1. Inspect hand tools and power tools for usability 2. Use hand tools properly and safely 3. Operate power tools properly and safely 4. Clean/maintain hand tools and power tools after use	12
SICIP-LE- MSP-04-S	Carry Out Measurements and Calculations	1. Check usability of measuring devices 2. Carry out accurate work measurements 3. Execute simple work calculations 4. Clean and maintain measuring instruments	12
Total Hour			40

OCCUPATION SPECIFIC (CORE) COMPETENCIES (290 HRS.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-MSP-01-O	Understand Mechanical Drawing	1. Interpret mechanical drawing 2. Create mechanical drawing	32
SICIP-LE-MSP-02-O	Perform Drilling Operations	1. Prepare drill machine 2. Perform drilling operations 3. Clean and store tools and equipment.	20
SICIP-LE-MSP-03-O	Perform Lathe Operations	1. Prepare lathe machine 2. Setup lathe works 3. Perform facing, straight, step, shoulder turning, knurling, grooving and parting-off operations. 4. Perform taper and eccentric turning. 5. Perform thread cutting operation 6. Clean and store tools and equipment.	90
SICIP-LE-MSP-04-O	Perform Milling Machine Operations	1. Prepare Milling machine. 2. Perform indexing operation using index head. 3. Carryout plain, side face, gang and straddle milling operations. 4. Carryout slot, key way, parting off, end, form and angular milling operations 5. Perform gear-cutting operation on milling machine. 6. Clean and store the tools and equipment.	100
SICIP-LE-MSP-05-O	Perform Shaper Machine Operations and Surface Grinding	1. Prepare shaper machine 2. Carryout shaping operations 3. Prepare for surface grinding machine 4. Perform surface grinding operation 5. Clean and store the tools and equipment.	48
Total Hours			290 hrs

COMPETENCY STANDARD: MACHINE SHOP PRACTICE

A. The Generic (Basic) Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-MSP-01-G
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Practice personal health and safety procedures	2.1 OHS policies and procedures are followed and practiced. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal health, hygiene and safety procedures are practiced.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. OHS policies	1.1 International/ Local OHS requirements 1.2 Fire Safety Rules and Regulations 1.3 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gloves 2.3 Safety shoes 2.4 Helmet 2.5 Face mask 2.6 Goggles and safety glasses 2.7 Ear plugs 2.8 Sun block 2.9 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards. 3.2 Biological hazards.

	3.3 Physical Hazards. 3.3.1 Machine hazards. 3.3.2 Materials hazards. 3.3.3 Tools and Equipment hazards.
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Applying CPR (Cardiopulmonary Resuscitation) 5.5 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures. 1.3 Emergency procedures. 1.3.1 Firefighting. 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of (biological, chemical and physical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 OHS awareness.
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Following personal work safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Using firefighting accessories and fire extinguishers 2.7 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Communication with peers, sub-ordinates and seniors in workplace 3.3 Promptness in carrying out activities 3.4 Tidiness and timeliness 3.5 Respect of peers, sub-ordinates and seniors in workplace 3.6 Environmental concern 3.7 Sincere and honest to duties
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Practiced personal health and safety procedures 1.4 Reported incidents arising from hazards and risks to authority.
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	1.5 Implemented plans and procedures to respond emergency 1.6 Applied basic first aid procedure.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Demonstration 2.3 Oral questioning 2.4 Interview
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: COMMUNICATE IN ENGLISH IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-MSP-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to communicate in English in the workplace. It specifically includes the tasks of reading, understanding and preparing documents in English, preparing workplace documents in English and performing conversations in English.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Read, understand and prepare documents in English	1.1 Workplace documents are read and understood 1.2 Visual information is interpreted. 1.3 Resume/CV is understood and prepared
2. Prepare workplace documents in English	2.1 Simple <u>routine workplace documents</u> are prepared using key words, phrases, simple sentences 2.2 <u>Visual aids</u> are prepared and submitted. 2.3 Key information is written in the appropriate places in standard forms.
3. Perform conversations in English language	4.1 Active listening is demonstrated. 4.2 Conversation is performed in English with peers, customers and management as per the workplace standard.

Range of Variables

Variable	Range
	May Include but not limited to:
1. Routine workplace documents	1.1 Meeting Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs
2. Visual aids	2.1 Maps 2.2 Diagrams 2.3 Forms 2.4 Labels 2.5 Graphs 2.6 Charts

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Interpret workplace documents in English 1.2 Preparing workplace documents in English 1.3 Listen to conversation in English 1.4 Job roles, responsibilities and compliances 1.5 Understanding of visual aids (e.g., graphs, presentations, or charts) and their preparation 1.6 Knowledge of workplace communication etiquette (e.g., formal vs. informal conversations).
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	1.7 Awareness of workplace standards for professional communication.
2. Underpinning Skills	2.1 Reading, Understanding, and Preparing Documents in English 2.2 Preparing Workplace Documents in English 2.3 Performing Conversations in English Language 2.4 Interpersonal skills to maintain polite and professional communication in workplace settings
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Promptness in carrying out activities 3.3 Tidiness and timeliness 3.4 Respect of peers, sub-ordinates and seniors in workplace 3.5 Environmental concern 3.6 Sincere and honest to duties
4. Resource Implications	The following resources must be provided: 4.1 Work place Procedure 4.2 Materials relevant to the proposed activity 4.3 All tools, equipment, material and documentation required. 4.4 Relevant specifications or work instructions

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Read, Understood, and Prepared Documents in English 1.2 Conversed in English with peers and customers. 1.3 Made reports of workplace documents in English.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Demonstration 2.3 Oral questioning 2.4 Interview
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: OPERATE IN A SELF-DIRECTED TEAM	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-MSP-03-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members.	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood, valued and practiced 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member.	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practice. 3.3 Team member's support to other members is appreciated and valued. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables

Variable	Range
	May Include but not limited to:
1. Forms of communication	1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective interpersonal skills to interact with team members 1.5 Effective formal and informal forms of communication 1.6 Value of diversity in team functioning. 1.7 Correct use of workplace terminology 1.8 Team's duties, responsibilities, authorities, objectives and task requirements 1.9 Support mechanism to other members of team to ensure achievements of goals. 1.10 Methods of identifying current and potential problems faced by a team 1.11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Identifying relationships within team and with other workers 2.4 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives 2.5 Using formal and informal forms of communication 2.6 Understanding and valuing views and opinions of other team members 2.7 Performing tasks in accordance with organizational and team requirements, and workplace procedures and practice. 2.8 Team member's support to other members is appreciated and valued 2.9 Identifying solutions to the problem 2.10 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Teamwork 3.2 Promptness in carrying out activities. 3.3 Tidiness and timeliness. 3.4 Respect of peers, sub-ordinates and seniors in workplace. 3.5 Sincere and honest to duties
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Pens 4.3 Papers 4.4 Work books 4.5 Learning manuals

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified team goals and work processes 1.2 Communicated and cooperated with team members. 1.3 Worked as a team member 1.4 Solved problems as a team member
2. Methods of Assessment	Methods of assessment may include but not limited to:

	2.1 Written test 2.2 Demonstration 2.3 Oral questioning
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

B. The Sector Specific (Common) Competencies

Unit of Competency: APPLY GREEN PRACTICES	Nominal Duration: 04 hrs.	Unit Code: SICIP-LE-MSP-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace, and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices</u> are explained. 1.2 <u>Sources of environmental impacts</u> during light engineering activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 <u>Ways of mitigating environmental impacts</u> in light engineering sector are explained.
2. Minimize resource use in the workplace	2.1 Water, energy, and raw material consumption are documented. 2.2 <u>Recyclable and non-recyclable items</u> are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of waste</u> are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair 1.3 use of sustainable materials with low environmental impact 1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of environmental impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical's leakage 2.3 Noise pollution

	2.4 Waste generation 2.4.1 waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of mitigating environmental impacts	3.1 Utilizing Energy-Efficient Equipment 3.2 Adopting Renewable Energy Sources 3.3 Implementing Site Protection Measures 3.4 Using Reusable Materials 3.5 Choosing Sustainable Materials 3.6 Using Noise-Reducing Equipment and scheduling work appropriately 3.7 Optimizing Logistics and Delivery
4. Recyclable and non-recyclable items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different types of waste	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Electrical system waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste 5.8 Chemical West 5.9 Fuel/Oil west

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in Light Engineering sector 1.2 Key terms and symbols related to environmental sustainability in Light Engineering drawings 1.3 Sources of environmental impacts in Light Engineering 1.4 Methods to minimize resource consumption (water, energy, raw materials) 1.5 Waste management practices 1.6 Sustainable alternatives to fossil-based energy resources 1.7 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	2.1 Identifying environmental impacts in Light Engineering activities 2.2 Applying methods to minimize resource use in the workplace 2.3 Implementing waste management practices 2.4 Following procedures to safely dispose of hazardous materials 2.5 Documenting water, energy, and material consumption in the workplace 2.6 Performing green practices in personal and professional activities
3. Underpinning Attitudes	3.1 Commitment to sustainable and eco-friendly Light Engineering

	practices 3.2 Active participation in reducing resource use and waste generation 3.3 Eagerness to adopt green habits in professional life 3.4 Promptness in implementing sustainable practices in Light Engineering activities 3.5 Tidiness and concern for environmental cleanliness 3.6 Compliance with safety standards and environmental regulations
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering hand tools and power tools 4.3 Safety gear 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Operation and maintenance manuals 4.8 Waster segregation bins

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Explained the principles of green practices in Light Engineering 1.2 Identified sources of environmental impacts 1.3 Implemented procedures to minimize resource use (water, energy, raw materials) 1.4 Disposed of hazardous waste in compliance with safety and environmental standards 1.5 Practiced green habits to reduce waste in both personal and professional life
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: TRANSLATE MANUALS, DRAWINGS, SPECIFICATIONS AND ESTIMATE	Nominal Duration: 12 hrs.	Unit Code: SICIP-LE-MSP-02-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to translate manuals, schematic drawings, specifications and work estimation. It specifically includes the tasks of understanding symbols, mechanical drawing, specifications and the way of estimation, interpreting manuals, drawings specific to Engineering works and storing manuals, drawings and plans.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand symbols, mechanical drawing, specifications and the way of estimation	1.1 Symbols are identified and interpreted 1.2 Drawing and specifications are identified 1.3 Way of estimation according to drawing and specifications is demonstrated
2. Interpret manuals, drawings specific to engineering works	2.1 <u>Manuals</u> and <u>mechanical drawings</u> are correctly recognized from manuals, designs and plans. 2.2 Terms and abbreviations in drawings are recognized. 2.3 <u>Signs and symbols</u> in the drawing are interpreted.
3. Store manuals, drawings and plans	3.1 Manuals, drawings, and plans are collected and packed. 3.2 Manuals, drawings and plans are stored to prevent damage, and ready access and updating of information when required.

Range of Variables

Variable	Range
	May include but not limited to:
1. Manuals	1.1 Manufacturer's Specification Manual 1.2 Repair Manual 1.3 Maintenance Procedure Manual 1.4 Periodic Maintenance Manual 1.5 Quality Manual 1.6 Instruction Manual
2. Mechanical Drawings	2.1 Drawings 2.2 Sketches
3. Signs and symbols	3.1 All signs and symbols associated with the mechanical components and devices

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Types of Manuals 1.2 Identification of Signs and Symbols 1.3 Interpretation of signs and symbols
2. Underpinning Skills	2.1 Checking version and date of the manual 2.2 Identifying relevant drawings and specifications correctly 2.3 Identifying terms and abbreviations 2.4 Identifying signs and symbols 2.5 Interpreting drawings and specifications 2.6 Storing manuals
3. Underpinning Attitudes	3.1 Eagerness to learn 3.2 Orderliness 3.3 Resourcefulness

4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering manuals and literatures 4.3 Pens 4.4 Papers 4.5 Work books
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Checked version and date of the manual 1.2 Identified relevant drawings and specifications correctly. 1.3 Identified terms and abbreviations. 1.4 identified Signs and symbols. 1.5 Interpreted drawings and specifications.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: WORK WITH HAND TOOLS AND POWER TOOLS	Nominal Duration: 12 hrs.	Unit Code: SICIP-LE-MSP-03-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to work with hand tools and power tools properly. It specifically includes the tasks of inspecting hand tools and power tools for usability, using hand tools properly and safely, operating power tools properly and safely and cleaning/maintaining hand tools and power tools after use.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Inspect hand tools and power tools for usability	1.1 Appropriate tools are selected. 1.2 Application of tools to job requirement is determined. 1.3 Usability of tools are checked and verified. 1.4 <u>Hand tools</u> and <u>power tools</u> are prepared. 1.5 Sources of power supply for power tools identified.
2. Use hand tools properly and safely	2.1 Appropriate hand tool for the job is used. 2.2 Proper and safe use/operation is applied in the different types of hand tools. 2.3 <u>Safety precautions</u> is observed when using hand tools. 2.4 Unsafe or faulty tools are identified and marked for repairing
3. Operate power tools properly and safely	3.1 Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.2 Proper sequence of operation is applied in using power tools to produce results. 3.3 Power tools are used safely in accordance to manufacturer's operating specification.
4. Clean/maintain hand tools and power tools after use	4.1 Dust and foreign materials are removed from power tools in accordance with workplace standard. 4.2 Scraps or wastages are removed and cleaned 4.3 Condition of tools is checked after use. 4.4 Appropriate lubricant is applied after use and prior to storage. 4.5 <u>Measuring tools</u> are checked and calibrated. 4.6 Defective tools, instruments, power tools and accessories are inspected and corrected or replaced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Hand tools	1.1 Screwdrivers 1.2 Pliers 1.3 Wire Strippers 1.4 Fish Wire (for pulling wire through conduit) 1.5 Leveler 1.6 Tester 1.7 Allen key 1.8 Nut Drivers 1.9 Crimping Tool (for wire terminals) 1.10 Flashlight or Headlamp 1.11 Hacksaw (for cutting conduit)

	1.12 Chisels and Hammer 1.13 Adjustable wrench 1.14 File 1.15 Socket wrench 1.16 Ratchet
2. Power tools	2.1 Power drills 2.2 Nail guns 2.3 Angle grinders and Grinders 2.4 Pneumatic wrenches 2.5 Pedestal drills
3. Safety precautions	3.1 Use of appropriate PPEs 3.2 Proper hand, feet and eye coordination 3.3 Working environment 3.4 Safe operating condition of hand tools and power tools 3.5 Awareness to OHS requirements
4. Measuring instruments	4.1 Tachometer 4.2 Vernier Calipers 4.3 Calipers 4.4 Micrometer 4.5 Bevel Protractor

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of tools, functions and use 1.2 Types of Hand tools and their proper use and techniques 1.3 Types of Power tools, use and safe handling method 1.4 Technical application of tools 1.5 Procedures in the use of hand tools and power tools 1.6 Policies and procedures for occupational health and safety 1.7 Use of PPE 1.8 Handling of tools and equipment 1.9 Reporting and documentation 1.10 Preventive maintenance 1.11 Methods and techniques 1.12 Quality procedures 1.13 Waste management 1.14 Storage procedures
2. Underpinning Skills	2.1 Using appropriate hand tool for the job 2.2 Observing safety precautions when using hand tools 2.3 Using power tools correctly and safely in accordance to manufacturer's operating specification. 2.4 Checking condition of tools after use 2.5 Applying appropriate lubricant on hand tools and power tools after use and prior to storage 2.6 Inspecting and correcting or replacing defective tools, instruments, power tools and accessories 2.7 Storing Tools and power tools safely in appropriate location
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Environmental concerns 3.3 Eagerness to learn 3.4 Tidiness and timeliness 3.5 Concern to proper use of tools 3.6 Orderliness
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering hand tools and power tools

	4.3 Pens 4.4 Papers 4.5 Work books 4.6 Operation and maintenance manuals
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Used appropriate hand tools for the job. 1.2 Observed safety precautions when using hand tools. 1.3 Used power tools safely in accordance to manufacturer's operating specification. 1.4 Cleaned and maintained hand tools and power tools after use and prior to storage. 1.5 Inspected and corrected or replaced defective tools, instruments, power tools and accessories. 1.6 Stored tools and power tools safely in appropriate location.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: CARRY OUT MEASUREMENTS AND CALCULATIONS	Nominal Duration: 12 hrs.	Unit Code: SICIP-LE-MSP-04-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry-out measurements and calculations. It specifically includes the tasks of checking usability of measuring devices, carrying out accurate work measurements, executing simple work calculations and cleaning and maintaining measuring instruments.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Check usability of measuring devices	1.1 Appropriate <u>measuring device</u> is selected for the job. 1.2 Applications of measuring device is determined. 1.3 Usability of measuring device is checked and verified. 1.4 Measuring device is prepared.
2. Carry out accurate work measurements	2.1 Measurements are obtained using appropriate measuring device. 2.2 <u>Systems of measurements</u> are identified and converted where necessary. 2.3 Results are confirmed and recorded.
3. Execute simple work calculations	3.1 Simple calculations involving <u>four basic mathematical calculations</u> are executed. 3.2 Other operations are used to complete tasks in Light Engineering works 3.3 Appropriate formulas for calculating quantities of materials are selected. 3.4 Calculations are performed and verified. 3.5 Material quantities are calculated. 3.6 Results are interpreted and communicated to authority.
4. Clean and maintain measuring instruments	4.1 Dust and foreign materials are removed from measuring instrument 4.2 Condition of instrument is checked 4.3 Appropriate lubricant is applied after use and prior to storage 4.4 Measuring instruments are checked and calibrated. 4.5 Instrument is stored in accordance to workplace procedure

Range of Variables

Variable	Range
	May include but not limited to:
1. Measuring device	1.1 Slide calipers 1.2 Steel tape 1.3 Steel rule 1.4 Tri-square 1.5 Feeler gauges 1.6 Spirit level 1.7 Dial Indicator 1.8 Bevel Protractor
2. Systems of measurements	2.1 Metric system 2.2 FPS system
3. Four basic mathematical Calculation	3.1 Area Calculation 3.2 Volume Calculation 3.3 Weight Calculation

	3.4 Angle Calculation
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Types and principles of operation of measuring devices 1.2 The ISO standard of measurements 1.3 Methods of measurement and calculation 1.4 Fraction and decimals 1.5 Linear measurement 1.6 Units of conversion and conversion factors in measurements 1.7 Dimensioning and fits and tolerances 1.8 Calculating ratio and proportion 1.9 Care in the use of measuring devices
2. Underpinning Skills	2.1 Selecting appropriate measuring device for the job 2.2 Checking and verifying usability of measuring device 2.3 Obtaining measurements using appropriate measuring device. 2.4 Confirming measurements and recording results 2.5 Carrying out simple calculations involving four basic mathematical operations 2.6 Calculating material quantities 2.7 Interpreting and communicating results to authority 2.8 Cleaning and storing measuring instruments
3. Underpinning Attitudes	3.1 Cleanliness/tidiness 3.2 Commitment to occupational health and safety practices 3.3 Environmental concerns 3.4 Eagerness to learn 3.5 Timeliness and orderliness 3.6 Respect for rights of peers and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of measuring and checking tools/instruments 4.3 Pens 4.4 Papers 4.5 Work books 4.6 Measuring tools' operating and maintenance manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Selected appropriate measuring device for the job. 1.2 Checked and verified usability of measuring device. 1.3 Obtained measurements using appropriate measuring device. 1.4 Confirmed measurements and recorded results. 1.5 Carried out Simple calculations involving four basic mathematical operations. 1.6 Calculated material quantities. 1.7 Interpreted and communicated Results to authority.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

C. Occupation-specific Competencies

Unit of Competency: UNDERSTAND MECHANICAL DRAWING	Nominal Duration: 32 hrs.	Unit Code: SICIP-LE-MSP-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand mechanical drawing. It specifically includes the tasks of interpreting mechanical drawing and creating mechanical drawing.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret mechanical drawing	1.1 Different types of <u>mechanical drawings</u> are interpreted. 1.2 Different views of mechanical drawing are identified and described. 1.3 Necessity of putting dimensions in different drawings are comprehended.
2. Create mechanical drawing	2.1 Drawing is created accurately. 2.2 Components, objects, parts, materials and assemblies are identified, drawn and described. 2.3 Base line or datum points are specified, as required. 2.4 Instructions are included in drawing. 2.5 Tolerance, limits and fits are identified in technical drawing.

Range of Variables

Variable	Range
	May include but not limited to:
1. Mechanical drawings	1.1 2D 1.1.1 Line 1.1.2 Rectangle 1.1.3 Circle 1.1.4 Ellipse 1.1.5 Polygon 1.1.6 Arc 1.2 3D 1.2.1 Block 1.2.2 Sphere 1.2.3 Pyramid 1.2.4 Taurus
2. Views of mechanical drawing	Drawing technique includes: 2.1 Top and Bottom View 2.2 Right and Left View 2.3 Front and View 2.4 Isometric View 2.5 Oblique View 2.6 Hidden view 2.7 Perspective view
3. Tolerance	3.1 General 3.2 Angular 3.3 Geometric

Circular Content Guide

1. Underpinning knowledge	1.1 Drawing tools and supplies 1.2 Drawing symbols 1.3 Drawing different Views 1.4 Dimensioning techniques 1.5 Tolerance, limits and fits 1.6 Engineering materials
2. Underpinning skills	2.1 Interpreting mechanical drawing 2.2 Creating mechanical drawing
3. Underpinning attitudes	3.1 Tidy and punctual 3.2 Prompt in carrying out activities 3.3 Sincere and honest concerning duties 3.4 Active on teamwork 3.5 Eager to learn 3.6 Concerned for proper use of tools 3.7 Concerned about the work environment 3.8 Committed to occupational health and safety practices
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Drawing tools and supplies 4.3 Specimen parts and components 4.4 Drawings and sketches 4.5 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 Interpreted mechanical drawing 1.2 Created mechanical drawing
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Oral test 2.3 Observation 2.4 Demonstration
3. Context of assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: PERFORM DRILLING OPERATIONS	Nominal Duration: 20 hrs.	Unit Code: SICIP-LE-MSP-2-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform drilling operation. It specifically includes the tasks of preparing drill machine, performing drilling operation, cleaning and storing tools and equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare drill machine	1.1 Different types of <u>drill machines</u> are identified. 1.2 Drill Machine is selected for different drilling operations. 1.3 <u>Major components</u> of drill machine is identified and interpreted. 1.4 Drilling machine mechanical feature, RPM, cutting speed and federate are demonstrated according to the machine specifications. 1.5 <u>Drill bits</u> and job materials are selected and collected according to the requirements of the operations. 1.6 Machine guards and coolant devices are checked in accordance with job requirement. 1.7 Work piece and drill bits are setup and clamped to required level of accuracy using instruments/equipment according to work site procedures. 1.8 Safe work practices are observed and personal protective equipment (PPEs) are worn.
2. Perform drilling operations	2.1 Work piece and drill bits are checked and clamped accurately. 2.2 Drill grinding parameters are interpreted. 2.3 Drill bits are grinded. 2.4 Drilling operations are performed as per the job requirements. 2.5 Job is checked/measured in conformance with specification
3. Clean and store tools and equipment.	3.1 Workplace, tools and equipment are cleaned and maintained in accordance with workplace requirements 3.2 Waste materials are disposed in proper place. 3.3 Tools, equipment and finished products are stored safely in accordance with workplace procedures

Range of Variables

Variable	Range May Include but not limited to:
1. Drill machines	1.1 Hand drill machine 1.2 Pedestal drill machine 1.3 Radial drill machine 1.4 Column drill machine
2. Major components	2.1 Drill head 2.2 Spindle axis 2.3 Chuck 2.4 Driving motor with control 2.5 Spindle lock 2.6 Work table 2.7 Column 2.8 Belt 2.9 Stepped pulley for speed changing

	2.10Base
3. Drill bits.	3.1 Twist drill bits 3.2 Wood or brad point bits 3.3 Masonry bit 3.4 Multi-purpose bit 3.5 Glass or tile bit 3.6 Countersink bit
4. Drill grinding parameters	4.1 Grinding Wheel speed 4.2 Feed rate 4.3 Grinding depth 4.4 Wheel rotation
5. Drilling operations	5.1 Spot Drilling 5.2 Straight hole drilling 5.3 Counter boring drilling 5.4 Countersinking 5.5 Reaming

Curricular Evidence Guide

1. Underpinning Knowledge	1.1 Types of Drilling Machine 1.2 Major components of Drill machine 1.3 Different parts and accessories of Drill Machine 1.4 Drill bits, job materials and specifications 1.5 Drilling operations 1.6 Drawing interpretation 1.7 Machine guards and coolant 1.8 Drill grinding parameters
2. Underpinning Skills	2.1 Setting drill machine 2.2 Setting work piece and drill bits 2.3 Selecting cutting speed (RPM), feed rate and depth of cut 2.4 Performing Drilling operation 2.5 Grinding drill bits in different angles 2.6 Checking and measuring ground drill using drill gauge.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.1 Communication with peers, sub-ordinates and seniors in workplace. 3.2 Tidiness and timeliness. 3.3 Environmental concern.
4. Resource Implications	4.1 The following resources MUST be provided: 4.2 Workplace. 4.3 Tools, equipment and facilities appropriate to processes or activity 4.4 Materials relevant to the proposed activity. 4.5 Outfits appropriate in applying safety measures. 4.6 Relevant drawings, manuals, codes, standards and reference material.

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Grinded drill bits 1.2 Set drill bit and workpiece . 1.3 Selected cutting speed (RPM), feed rate 1.4 Interpreted component drawing and identified specifications 1.5 Performed drilling operations 1.6 Checked and measured job in conformance to specification
2. Methods of	Competency should be assessed by:

Assessment	2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be finished in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: PERFORM LATHE OPERATIONS	Nominal Duration: 90 hrs.	Unit Code: SICIP-LE-MSP-3-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform lathe operations. It specifically includes the tasks of preparing lathe machine, setting up lathe works, performing facing, straight turning, step turning, shoulder turning, Knurling, grooving and parting-off operation and performing taper, eccentric turning, performing thread cutting operation, and cleaning and storing tools and equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare lathe machine	1.1 Different <u>types of lathe machines</u> are identified. 1.2 Lathe machine is selected for lathe operations. 1.3 <u>Major parts of lathe machine</u> are identified and interpreted. 1.4 <u>Lathe accessories</u> are used appropriately to the requirements of the operations. 1.5 Drawings are interpreted to produce component in accordance with the job specifications. 1.6 Job materials are selected and collected according to the job specifications. 1.7 Cutting tool is selected in accordance with the requirements of the operation. 1.8 Sequence of operation is determined to produce component in accordance with required specifications 1.9 PPEs are selected and used. 1.10 Safe work practices are maintained and personal protective equipment (PPE) are worn as required for the job performed.
2. Setup lathe works	2.1 Grinding of tool blank following the profile angles of Lathe cutting tools is performed. 2.2 Work piece is centered and clamped on chuck to required level of accuracy using tools and equipment in accordance with work procedures. 2.3 Work piece is setup and clamped to required level of accuracy using instruments/equipment according to work procedures. 2.4 Cutting tool is set in accordance with the requirement of the operation. 2.5 Lathe accessories are used appropriately to the requirements of the jobs. 2.6 Machine guards and coolant devices are checked according to work requirement.
3. Perform facing, straight, step, shoulder turning, knurling, grooving and parting-off operations.	3.1 Work piece is clamped and dialed 3.2 Cutting speed, RPM, feed rate and depth of cut are calculated and adjusted as per job requirement. 3.3 Machine performance is checked in accordance with the job requirement. 3.4 Coolant is applied to prevent over heating of work piece and cutting tool. 3.5 Straight, step and shoulder turning are performed 3.6 Grooving operation is performed 3.7 Parting-off operation is performed 3.8 Knurling operation is performed 3.9 Job is checked/measured in conforming to specification using appropriate techniques, measuring tools and equipment.

4. Perform taper and eccentric turning.	4.1 Machine performance is checked in line with the job requirement. 4.2 Cutting speed, RPM, feed rate and depth of cut are calculated and adjusted. 4.3 Coolant is applied to prevent over heating of work piece and cutting tool. 4.4 Taper turning methods are selected in accordance with the job specifications. 4.5 Taper turning operation is performed in accordance with the job specifications. 4.6 Eccentric turning method is selected in accordance with the job specifications. 4.7 Eccentric turning operation is performed in accordance with the job specifications. 4.8 Job is checked and measured in conforming to the specification.
5. Perform thread cutting operation	5.1 Cutting speed, RPM, feed rate and depth of cut are calculated and adjusted as per job requirement. 5.2 Different types of thread are cut in accordance with the specifications outlined in the drawing. 5.3 Machine performance is checked in accordance with the job requirement. 5.4 Coolant is applied to prevent over heating of work piece and cutting tool. 5.5 External and internal V-threads are cut in accordance with specifications in the drawing. 5.6 External and internal ACME threads are cut in accordance with the specifications in the drawing. 5.7 Square-threads are cut in accordance with the specifications in the drawing. 5.8 Job is checked and measured in accordance with by using appropriate techniques, measuring tools and equipment.
6. Clean and store tools and equipment.	6.1 Workplace, Tools, equipment are cleaned and maintained in accordance with workplace requirements. 6.2 Preventive maintenance schedules are applied in accordance with workplace requirement. 6.3 Waste materials are disposed in proper place. 6.4 Tools, equipment and finished products are stored safely in accordance with workplace procedures

Range of Variables

Variable	Range May Include but not limited to:
1. Types of lathe machine.	1.1 Center lathe 1.2 Turret lathe 1.3 Tool room lathe 1.4 Capstan lathe 1.5 Multi spindle lathe 1.6 CNC lathe
2. Major parts of lathe machine	2.1 Headstock 2.2 Tailstock 2.3 Bed 2.4 Chuck 2.5 Carriage.

	2.6 Lead screw 2.7 Tool post 2.8 Compound rest Cross slide
3. Lathe accessories	3.1 Steady rest 3.2 Follower rest 3.3 Feed dog 3.4 Face Plate 3.5 Dog plate 3.6 Collet 3.7 Mandrels 3.8 Angle plate
4. Profile angles	4.1 Back/rake angle. 4.2 Front clearance angle. 4.3 Side rake angle 4.4 Side clearance angle. 4.5 End cutting-edge angle. 4.6 Side cutting-edge angle.
5. Taper turning methods.	5.1 Form tool methods 5.2 Compound slide methods. 5.3 Setting over the tailstock. 5.4 Taper turning with attachment.
6. Eccentric turning methods	6.1 Using 3-jaw universal chuck packing a piece to offset. 6.2 Using 4-jaw independent chuck
7. Types of thread	7.1 British standard whit worth threads. Thread angle- 55 degrees. 7.2 Metric threads. Thread angles- 60 degrees. 7.3 Acme threads. Thread angle- 30 degrees. 7.4 Square threads. Thread angle- 90 degrees.
8. Preventive maintenance	8.1 Oil and grease the machine sliding parts, daily, weekly and monthly. 8.2 Coolant tank schedule cleaning. 8.3 Electrical fixture & connections regularly Checkup. 8.4 Cleaning the table and vise after each operation.

Curricular Evidence Guide

1. Underpinning Knowledge	1.1 Types of lathe machine and its major components 1.2 Lathe machine accessories 1.3 Cutting speed, feed rate and depth of cut 1.4 Job materials and cutting tools 1.5 Profile angle of cutting tools 1.6 Facing, straight and step turning, shoulder turning, knurling, grooving and parting-off 1.7 Taper and eccentric turning 1.8 Threading
2. Underpinning Skills	2.1 Performing Profile angle of cutting tools 2.2 Performing facing, straight and step turning, shoulder turning, knurling, grooving and parting-off 2.3 Performing Taper and eccentric turning operation 2.4 Performing Thread operation
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Communication with peers, sub-ordinates and seniors in workplace. 3.3 Promptness in carrying out activities. 3.4 Tidiness and timeliness. 3.5 Respect of peers, sub-ordinates and seniors in workplace.

	3.6 Environmental concern.
4. Resource Implications	The following resources MUST be provided: 4.1 Workplace. 4.2 Tools, equipment and facilities appropriate to processes or activity 4.3 Materials relevant to the proposed activity. 4.4 Equipment and outfits appropriate in applying safety measures. 4.5 Relevant drawings, manuals, codes, standards and reference material.

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Performed straight, step and shoulder turning after facing in accordance with specifications in the drawing 1.2 Performed knurling operation 1.3 Performed grooving operation after turning in accordance with the specifications in the drawing 1.4 Performed parting-off operation and produced job in accordance with specification in the drawing. 1.5 Performed taper turning operation using form tool, compound slide, offsetting tailstock and taper turning attachment to produce component in accordance with specifications in the drawing. 1.6 Performed eccentric turning operation in accordance with specifications in the drawing. 1.7 Performed external and internal V-thread cutting in accordance with specifications in the drawing. 1.8 Performed external and internal ACME cutting in accordance with specifications in the drawing. 1.9 Performed square-threads cutting in accordance with specifications in the drawing.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be finished in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: PERFORM MILLING MACHINE OPERATIONS	Nominal Duration: 100hrs.	Unit Code: SICIP-LE-MSP-4-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform milling machine operation. It specifically includes the tasks of preparing milling operations, performing indexing operation using index head, carrying out plain, side, face, gang and straddle milling operations, carrying out slot, key way, parting off, end, form and angular milling operations, performing gear cutting operation on milling machine and cleaning and storing the tools and equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare Milling machine.	1.1 Different <u>types of Milling machines</u> are identified. 1.2 Milling machine is selected for milling operations. 1.3 <u>Major parts of Milling machine</u> are identified and interpreted. 1.4 <u>Milling accessories</u> are used appropriately as per requirements of the operations. 1.5 Drawings are interpreted to produce component in accordance with the job specifications. 1.6 Job materials are selected and collected according to the job specifications. 1.7 Cutting tool is selected in accordance with the requirements of the operation. 1.8 Sequence of operation is determined to produce component in accordance with required specifications 1.9 PPEs are selected and used. 1.10 Safe work practices are maintained and personal protective equipment (PPE) are worn as required for the job performed.
2. Perform indexing operation using index head.	2.1 Index head is selected, collected and checked 2.2 Different parts of index head are identified, checked and tested 2.3 Index head is set on milling machine in accordance with instruction of manual. 2.4 Work piece is mounted in the index head properly. 2.5 <u>Different types of indexing methods</u> are identified and calculated in accordance with identified indexing formula 2.6 Different indexing methods are performed
3. Carryout plain, side face, gang and straddle milling operations.	3.1 Required material and <u>milling cutters</u> are selected and set as per job requirement. 3.2 Drawings and specification for performing plain, side face, gang and straddle operation are interpreted. 3.3 <u>Cutting parameters</u> of milling machine are identified. 3.4 Sequence of operation is determined to perform milling work according to specifications. 3.5 Machine performance is checked 3.6 Plain, side, face, gang and straddle milling operation are performed 3.7 Job is checked/measured in line with the specifications.
4. Carryout slot, key way, parting off, end, form and angular milling operations.	4.1 Drawings and specification are interpreted in relation to slot, key way, parting off, end, form and angular milling operation. 4.2 Sequence of operation is determined to perform milling work. 4.3 Machine performance is checked. 4.4 Slot, key way, parting off, end, form and angular milling operation are performed. 4.5 Job is checked/measured according to specification.

5. Perform gear-cutting operation on milling machine.	5.1 Drawings and specification are interpreted in relation to different gear cutting milling operation. 5.2 Gear cutter is selected and coolant is used. 5.3 Indexing is determined 5.4 Machine performance is checked in accordance with the job requirement 5.5 Different types of gears are interpreted. 5.6 Gear cutting operation is performed as per gear module. 5.7 Job is checked/measured according to specification.
6. Clean and store the tools and equipment.	6.1 Workplace, tools, equipment and milling machine are cleaned. 6.2 Preventive maintenance schedules are applied 6.3 Waste materials are disposed of proper place. 6.4 Tools, equipment and finished products are stored safely in appropriate location

Range of Variables

Variable	Range May include but not limited to:
1. Types of milling machine.	1.1 Horizontal milling machine 1.2 Vertical milling machine 1.3 Universal milling machine. 1.4 CNC milling machine
2. Major components of Milling machine	2.1 Base 2.2 Column 2.3 Knee 2.4 Table 2.5 Arbor 2.6 Saddle 2.7 Spindle
3. Milling accessories	3.1 Machine vice. 3.2 Swivel base. 3.3 Angle plate. 3.4 Dividing plate. 3.5 3-Jaw universal chuck. 3.6 Collet chuck. 3.7 Rotary table 3.8 Indexing head
4. Different types of indexing methods	4.1 Simple indexing 4.2 Compound indexing 4.3 Differential indexing
5. Milling cutters.	5.1 Side and face cutter. 5.2 Slotting cutter. 5.3 Double angle cutter. 5.4 End mill cutter. 5.5 Corner radius cutter. 5.6 Shell end mill cutter. 5.7 Face milling cutter. 5.8 Involute gear cutter. 5.9 Gear hob cutter. 5.10 Slab milling. 5.11 T-slot cutter. 5.12 Ball nose cutter.
6. Cutting parameters	6.1 Cutting speed 6.2 Feed

	6.3 Depth of cut
7. Different types of gear.	7.1 Spur gear 7.2 Helical or spiral gear 7.3 Bevel gear 7.4 Worm and worm gear 7.5 Rack and pinion

Curricular Evidence Guide

1. Underpinning Knowledge	1.1 Types of milling machines and their major components 1.2 Appropriate use of milling accessories. 1.3 Cutting speed, feed rate and depth of cut 1.4 Job materials and milling cutters 1.5 Indexing 1.6 Plain, side face, gang and straddle milling operations 1.7 slot, key way, parting off, end, form and angular milling operations 1.8 gear-cutting operation 1.9 Operating parameters of milling machine
2. Underpinning Skills	2.1 Performing different indexing. 2.2 Performing plain, side, face, gang and straddle milling operation. 2.3 Performing slot, key way, parting off, end, form and angular milling operation. 2.4 Performing different types of gear cutting operations. 2.5 Checking and measuring of work piece in conformance with specification using appropriate techniques, measuring tools and equipment. 2.6 Applying preventive maintenance schedules in accordance to the machine manufacturer's requirement.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Communication with peers, sub-ordinates and seniors in workplace. 3.3 Promptness in carrying out activities. 3.4 Tidiness and timeliness. 3.5 Respect of peers, sub-ordinates and seniors in workplace. 3.6 Environmental concern.
4. Resource Implications	The following resources must be provided: 4.1 Workplace 4.2 appropriate Tools, equipment and facilities 4.3 Materials relevant to the proposed activity. 4.4 Equipment and outfits appropriate in applying safety measures. 4.5 Relevant drawings, manuals, codes, standards and reference material.

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Performed different indexing methods in accordance with job requirement specifications. 1.2 Performed plain, side, face, gang and straddle milling operation in accordance with the job requirement. 1.3 Performed Slot, key way, parting off, end, form and angular milling operation in accordance with the job requirement. 1.4 Performed different types of gear cutting operations according to the job requirement. 1.5 Checked and measured job in conformance to specification using appropriate techniques, measuring tools and equipment.
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	1.6 Applied Preventive maintenance schedules in accordance with machine manufacturer's requirements.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be finished in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: PERFORM SHAPER MACHINE OPERATIONS AND SURFACE GRINDING	Nominal Duration: 48 hrs.	Unit Code: SICIP-LE-MSP-5-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required of a machinist to perform shaper machine operation. It specifically includes the tasks of preparing shaper machine, carrying out shaping operations, preparing surface grinding machine, performing surface grinding operation and cleaning and storing the tools and equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare shaper machine	1.1 Different <u>types of Shaper machines</u> are identified. 1.2 Shaper machine is selected for shaper operations. 1.3 <u>Major parts of Shaper machine</u> are identified and interpreted. 1.4 <u>Shaper accessories</u> are used appropriately to the requirements of the operations. 1.5 Drawings are interpreted to produce component in accordance with the job specifications 1.6 Shaper machine function, quick return mechanism, principle are interpreted. 1.7 Feed rate and stroke length is selected in accordance with the job specifications. 1.8 Safe work practices are maintained and personal protective equipment (PPEs) are worn in accordance with workplace requirements.
2. Carryout shaping operations	2.1 Drawings and specification are interpreted in relation to the shaping operation. 2.2 Sequence of operation in shaping work is determined in accordance with specifications. 2.3 Machine performance is checked in accordance with job requirement. 2.4 <u>Shaping operations</u> are performed in accordance with the job requirement. 2.5 Job is checked and measured in conformance with specification using appropriate techniques, measuring tools and equipment.
3. Prepare surface grinding machine	3.1. Different <u>types of surface grinding machines</u> are identified. 3.2. Surface grinding machine is selected for grinding operations. 3.3. <u>Major parts of surface grinding machine</u> are identified and interpreted. 3.4. <u>Work holding device</u> are set appropriately to the requirements of the operations 3.5. Grinding wheel is selected and installed
4. Perform surface grinding operation	4.1. Grinding wheels are balanced, and dressed in accordance with the job requirement. 4.2. Work piece is set on the machine vice/magnetic vice. 4.3. RPM, cutting speed, feed rate and depth of cut are calculated as per job requirement 4.4. Machine performance is checked in conformance with the job requirement. 4.5. Coolant is applied to prevent over heating of the work piece and grinding wheel. 4.6. Surface grinding operation is performed in accordance with workplace requirement.

	4.7. Job is checked and measured in conformance with specification and appropriate techniques, measuring tools, and equipment are used
5. Clean and store the tools and equipment.	5.1. Workplace, tools, equipment and shaper machine are cleaned. 5.2. Preventive maintenance schedules are applied in accordance to workplace requirement. 5.3. Waste materials are disposed of proper place. 5.4. Tools, equipment and finished products are stored safely in appropriate location.

Range of Variables

Variable	Range May include but not limited to:
1. Shaper machine types	1.1 Horizontal type 1.2 Vertical type 1.3 Universal type
2. Major parts of shaper machine.	2.1. Ram 2.2. Tool head 2.3. Base 2.4. Column 2.5. Table 2.6. Saddle 2.7. Cross rail 2.8. Quick return mechanism
3. Shaper accessories	3.1. Machine vice 3.2. Angle plate 3.3. Index plate 3.4. Clamp 3.5. T – bolt 3.6. V-block
4. Principle of shaping operation	4.1. The arm is in reciprocating motion to push cutting tool back and forward across the work piece. 4.2. Cutting stroke arm move 220 degrees 4.3. Return stroke arm quick return to 140 degrees
5. Shaping operations	5.1. Horizontal plain surface 5.2. Vertical plain surface 5.3. Inclined/Angular surface 5.4. Groove surface 5.5. Slot surface 5.6. Blind hole, key way, spline and gear teeth.
6. Types of surface grinding machines	6.1. Horizontal 6.2. Vertical 6.3. Rotary
7. Major parts of surface grinding machine	7.1. Base 7.2. Column 7.3. Spindle 7.4. Grinding wheel 7.5. Table
8. Work holding device	8.1. Machine Vice 8.2. Magnetic chuck 8.3. Fixture
9. Preventive maintenance	9.1. Oiling and greasing the machine sliding parts on daily, weekly or monthly basis 9.2. Cleaning coolant tank 9.3. Checking up electrical fixture & connections regularly 9.4. Cleaning the table and vice after each operation.

Curricular Evidence Guide

1. Underpinning Knowledge	1.1 Types of Shaper machines and their major components 1.2 Appropriate use of Shaper accessories 1.3 Feed rate and stroke length 1.4 Shaping operations principle 1.5 Quick return mechanism 1.6 Types of surface grinding machines and their major components 1.7 Wheel dressing 1.8 Surface grinding operation principle
2. Underpinning Skills	2.1 Determining sequence of operation to produce component to the specifications of requirement. 2.2 Grinding of tool blank in accordance with the profile angles of shaping single point cutting tool 2.3 Performing Shaper machine operation 2.4 Performing Wheel dressing 2.5 Performing surface grinding operation 2.6 Checking and measuring the job in conformance with specification and using appropriate techniques, measuring tools and equipment
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Communication with peers, sub-ordinates and seniors in workplace. 3.3 Promptness in carrying out activities. 3.4 Tidiness and timeliness. 3.5 Respect of peers, sub-ordinates and seniors in workplace. 3.6 Environmental concern.
4. Resource Implications	The following resources must be provided: 4.1 Workplace 4.2 Appropriate Tools, equipment and facilities 4.3 Materials relevant to the proposed activity. 4.4 Equipment and outfits appropriate in applying safety measures. 4.5 Relevant drawings, manuals, codes, standards and reference material.

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Performed Shaper machine operation 1.2 Performed Wheel dressing 1.3 Performed surface grinding operation 1.4 Checked and measured job in conformance to specification using appropriate techniques, measuring tools and equipment.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be finished in a training center or in an actual or simulated work place after completion of the training module.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Machine Shop Practice
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom and Workshop/ Lab):

- Classroom : 350 sft
- Workshop/ Lab : 800-1000 sft

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Lathe Machine	10
2.	Shaper Machine	1
3.	Milling Machine	1
4.	Drill Machine	2
5.	Bench Vice	8
6.	Surface Grinding Machine	2
7.	Power hacksaw	1
8.	Cut off saw	1
9.	Working Table	5

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP. For the operation of training course on Machine Shop Practice, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.